



Cambridge IGCSE™

MATHEMATICS

0580/22

Paper 2 (Extended)

March 2020

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the March 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **7** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

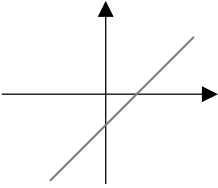
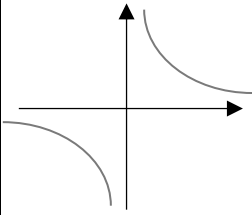
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

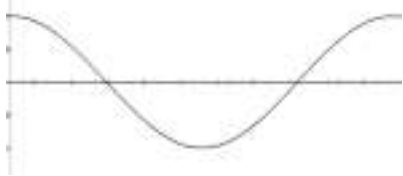
Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	12	1	
1(b)	8	1	
1(c)	5	1	
1(d)	$\sqrt{7}$	1	
2(a)	$\begin{array}{c ccc} 1 & 3 & 5 & 8 \\ 2 & 1 & 4 & 6 & 8 \\ 3 & 6 & 8 & 8 \\ 4 & 5 & 8 \end{array}$	2	M1 for correct but not ordered or for two correct rows ordered
2(b)	27	1	
3	$\begin{pmatrix} -4 \\ 3 \end{pmatrix}$	1	
4	165	2	M1 for $\frac{(24-2) \times 180}{24}$ or $180 - \frac{360}{24}$
5	$\frac{15}{28} \times \frac{7}{4}$ or $\frac{15}{28} \div \frac{16}{28}$ oe	M1	
	$\frac{15}{16}$ cao	A2	A1 for $\frac{105}{112}$ oe
6	7.2	3	M1 for $5 \times 8 + 6 \times 5 + 7 \times 11 + 8 \times 7 + 9 \times 5 + 10 \times 4$ M1dep for $\div 40$
7(a)	45.9	2	M1 for $0.5 \times 8.5 \times 10.8$ oe
7(b)	33[.0] or 33.04...	3	M2 for $8.5 + 10.8 + \sqrt{8.5^2 + 10.8^2}$ oe or M1 for $8.5^2 + 10.8^2$ oe
8	2.98×10^{-3}	1	
9(a)	$3x(x - 4y)$ final answer	2	B1 for $3(x^2 - 4xy)$ or $x(3x - 12y)$

Question	Answer	Marks	Partial Marks
9(b)	$m^2 - m - 6$ final answer	2	M1 for 3 terms from $m^2, -3m, +2m, -6$
10(a)	Correct sketch 	1	Line with positive gradient and negative y intercept
10(b)	Correct sketch 	2	B1 for only one branch or attempt at correct shape
11(a)	Rotation 90° clockwise oe (0, 2)	3	B1 for each
11(b)	Reflection $y = x$	2	B1 for each
11(c)	Enlargement [sf] $\frac{1}{2}$ (4, 6)	3	B1 for each
12	229 500 cao	3	B2 for 229 460... OR M1 for $250\,000 \times \left(1 - \frac{1.7}{100}\right)^5$ oe B1 for <i>their</i> more accurate answer correctly rounded to the nearest 100
13	$2.\dot{6} - 0.2\dot{6}$ oe	M1	
	$\frac{4}{15}$ oe fraction nfw	A1	If M0 scored SC1 for $\frac{k}{90}$
14(a)	11.5	1	
14(b)(i)	12	1	
14(b)(ii)	8.5	1	

Question	Answer	Marks	Partial Marks
15	116°	B1	
	alternate segment theorem	B1	
	angles in opposite segments are supplementary or cyclic quadrilateral or angles at a point on a straight line	B1	
16	$8y^2 - 42y + 10 [= 0]$ or $8x^2 + 14x - 400 [= 0]$	M3	M1 for $(7 - 3y)^2 - y^2 = 39$ oe or $x^2 - \left(\frac{7-x}{3}\right)^2 = 39$ oe M1 for $49 - 21y - 21y + 9y^2$ or better or $49 - 7x - 7x + x^2$ or better or for correct expansion of their quadratic binomial
	$(8y - 2)(y - 5) [= 0]$ oe $(8x - 50)(x + 8) [= 0]$ oe	M1	M1 for correct method to solve <i>their</i> quadratic equation e.g. factors, quadratic formula, completing the square
	$x = 6.25$ oe $y = 0.25$ oe $x = -8$ $y = 5$	B2	B1 for $x = 6.25$, $x = -8$ or for $y = 0.25$, $y = 5$ or for a correct pair of x and y values
17	$[y =] -\frac{1}{6}x + \frac{11}{2}$ oe	4	M1 for [gradient of $AB =] \frac{5 - -7}{3 - 1}$ oe M1 for [gradient of perpendicular =] $-\frac{1}{\text{their grad } AB}$ M1 for substituting (3, 5) in <i>their</i> linear equation
18	22.5 nfww	3	M2 for $\frac{146.2 + 0.05}{7 - 0.5}$ or M1 for $146.2 + 0.05$ or $7 - 0.5$ or better seen
19(a)	Correct sketch 	2	Needs all three features for 2 marks: • Correct curve shape • Maximum at (0, 1) and at (360, 1) and minimum at (180, -1) • Passing through (90, 0) and (270, 0) only B1 for two correct features

Question	Answer	Marks	Partial Marks
19(b)	75.5 or 75.52... and 284.4 to 284.5	3	B2 for one correct or M1 for $\cos x = \frac{1}{4}$ oe If 0 scored, SC1 for two answers with a sum of 360
20	$[a =] 36$ $[b =] -6$	2	B1 for each or SC1 for correct answers reversed
21	X, Y and Z are collinear oe	1	Allow in a straight line
	X is the midpoint of ZY oe	1	Allow e.g. $ZY = 2XY$, $ZX = XY$ oe